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# CLASS: P5
# ROLL NO. : 17

WEEK-8 C LAB PROGRAMS

# PROGRAM-1

## CODE

```c
//PES1UG25AM095


#include<stdio.h>

int main()
{
char str1[]="programming";
int count=0;
char *ptr1;
ptr1=str1;
char max_character;
int max_count_of_character=0;

printf("The string is: %s\n",str1);
while(*ptr1!='\0')
{
count++;
ptr1++;
}
printf("Number of characters in string is : %d\n",count);

for(int i=0;i<count;i++)
{int freq=1;
    for(int j=i+1;j<count;j++)
    {
        if(str1[i]==str1[j])
        {
            freq++;
        }

    }
    if(freq>max_count_of_character)
    {
        max_count_of_character=freq;
        max_character=str1[i];
    }
}

printf("The max_character is %c\n",max_character);
printf("The number of times %c appeared is %d",max_character,max_count_of_character);

return 0;
}
```

## OUTPUT

```
The string is: programming
Number of characters in string is : 11
The max_character is r
The number of times r appeared is 2
PS C:\Users\Chethan S\Documents\WEEK-8>
```

# PROGRAM-2

## CODE

```c
//PES1UG25AM095

#include<stdio.h>
#include<string.h>
int main()
{
char first_name[50],last_name[50],full_name[100];

printf("Enter the first name of student: ");
scanf("%s",first_name);


printf("Enter the last name of the student: ");
scanf("%s",last_name);

strcpy(full_name,first_name);
strcat(full_name," ");
strcat(full_name,last_name);

printf("The full name of student is : %s\n",full_name);
printf("Length of full name is: %d\n",strlen(full_name));
char copied_name[100];
strcpy(copied_name,full_name);

printf("The copied name of student is : %s\n",copied_name);

char another_full_name[100];

printf("Enter another name of student: ");
getchar();
scanf("%[^\n]", another_full_name);

if(strcmp(copied_name,another_full_name)==0)
{printf("Both names are same");}
else
{printf("Both names are different");}

}
```

## OUTPUT

```
Enter the first name of student: PES
Enter the last name of the student: UNIVERSITY
The full name of student is : PES UNIVERSITY
Length of full name is: 14
The copied name of student is : PES UNIVERSITY
Enter another name of student: PESIT BANGLORE
Both names are different
PS C:\Users\Chethan S\Documents\WEEK-8>
```

# PROGRAM-3

## CODE

```c
//PES1UG25AM095
#include<stdio.h>

int main()
{
    char password[]="PES123";
    char input_password[50];
    int flag=0;
    printf("Enter the password: ");
    scanf("%s",input_password);


    char *ptr1,*ptr2;
    ptr1=password;
    ptr2=input_password;

    while(*ptr1!='\0' || *ptr2!='\0')
    {
        if(*ptr1==*ptr2)
        {
            flag=1;
        }
        else
        {printf("PASSWORD did not match . Acess denied.");
         return 0;
        }
        ptr1++;
        ptr2++;
    }

    if(*ptr1=='\0' && *ptr2!='\0' || *ptr1!='\0' && *ptr2=='\0')
    printf("PASSWORD did not match . Acess denied.");
    else if(flag==0)
    {printf("PASSWORD did not match . Acess denied.");}
    else if(flag==1)
    {printf("PASSWORD did matches . Acess granted.");}

}
```

## OUTPUT

```
ts\WEEK-8\" ; if ($?) { gcc 3.c -o 3 } ; if ($?) { .\3 }
Enter the password: PESU
PASSWORD did not match . Acess denied.
PS C:\Users\Chethan S\Documents\WEEK-8> cd "c:\Users\Chethan S
ts\WEEK-8\" ; if ($?) { gcc 3.c -o 3 } ; if ($?) { .\3 }
Enter the password: PES123
PASSWORD did matches . Acess granted.
PS C:\Users\Chethan S\Documents\WEEK-8>
```

# PROGRAM-4

## CODE

```c
//PES1UG25AM095
#include<stdio.h>

int main()
{
char first_name[50],last_name[50],full_name[100];
char *ptr1,*ptr2,*ptr3;
int count=0;
ptr3=full_name;
printf("Enter the first name of student: ");
scanf("%s",first_name);
ptr1=first_name;


printf("Enter the last name of the student: ");
scanf("%s",last_name);
ptr2=last_name;

while(*ptr1!='\0')
{
    *ptr3=*ptr1;
    ptr1++;
    ptr3++;
    count++;

}

*ptr3=' ';
ptr3++;
count++;

while(*ptr2!='\0')
{
    *ptr3=*ptr2;
    ptr2++;
    ptr3++;
    count++;

}
*ptr3='\0';
printf("%s\n",full_name);
printf("Number of characters in the full name is %d",count);
}
```

## OUTPUT

```
ts\WEEK-8\" ; if ($?) { gcc 4.c -o 4 } ; if ($?) {
Enter the first name of student: PES
Enter the last name of the student: UNIVERSITY
PES UNIVERSITY
Number of characters in the full name is 14
PS C:\Users\Chethan S\Documents\WEEK-8>
```

# CODE

```c
//PES1UG25AM095
#include <stdio.h>
#include <stdlib.h>

int main() {
    int size,new_size;
    int *malloc_array, *calloc_array;

    printf("Enter number of elements: ");
    scanf("%d", &size);

    malloc_array = (int*)malloc(size * sizeof(int));
    calloc_array = (int*)calloc(size, sizeof(int));

    printf("Enter %d elements for malloc array:\n", size);
    for (int i = 0; i < size; i++) {
        scanf("%d", &malloc_array[i]);
    }

    printf("Elements stored in malloc array:\n");
    for (int i = 0; i < size; i++) {
        printf("%d ", malloc_array[i]);
    }
    printf("\n");

    printf("Elements stored in calloc array:\n");
    for (int i = 0; i < size; i++) {
        printf("%d ", calloc_array[i]);
    }
    printf("\n");

    printf("Enter new size for realloc: ");
    scanf("%d", &new_size);

    malloc_array = (int*)realloc(malloc_array, new_size * sizeof(int));

    printf("Enter %d elements after realloc:\n", new_size);
    for (int i = 0; i < new_size; i++) {
        scanf("%d", &malloc_array[i]);
    }

    printf("Elements after realloc:\n");
    for (int i = 0; i < new_size; i++) {
        printf("%d ", malloc_array\
            [i]);
    }
    printf("\n");

    free(malloc_array);
    free(calloc_array);

    printf("Memory freed successfully.\n");

    return 0;
}
```

# PROGRAM-5

## OUTPUT

```
ts\WEEK-8\" ; if ($?) { gcc 5.c -o 5 } ; if ($?)
Enter number of elements: 5
Enter 5 elements for malloc array:
10 20 30 40 50
Elements stored in malloc array:
10 20 30 40 50
Elements stored in calloc array:
0 0 0 0 0
Enter new size for realloc: 3
Enter 3 elements after realloc:
90 80 70
Elements after realloc:
90 80 70
Memory freed successfully.
PS C:\Users\Chethan S\Documents\WEEK-8> \
```